Secret messages: encryption basics

Cyber security · Level 3 · Lesson 4



In this lesson you will

- Explain encryption, keys and decryption.
- Use Caesar’s cipher, and crack it by trying keys or counting letters.
- Explain why modern encryption is strong, and where it protects you.

Keys are everything

- Caesar: shift every letter; the shift is the key
- Few keys (25 or 27): try them all
- Al-Kindi: count the letters
- Modern keys: too many to try, so keep them secret

Cipher challenge

1. Make (4 min): Build a cipher strip from two alphabets.
2. Hide (5 min): Encrypt a message with a secret shift.
3. Crack (8 min): Swap and crack each other’s message.
4. Compare (3 min): Which method was faster, and why?

Arabic messages: use the 28 letters in order (ا ب ت ث ...), and write آ إ أ as ا, ؤ as و, ئ and ى as ي, and ة as ه, as the Labs app does.

Think about it

1. Why do long messages make Caesar’s cipher easier to crack?
2. If the method is public, what keeps a message secret?
3. Why do attackers trick people instead of breaking encryption?

Before you go

- One place encryption protects me + one key I’ll keep to myself.